Supplementary information

Disease-blind reconstruction distinguishes reproducible interferon remodeling from less stable B-cell state assignments in systemic lupus erythematosus

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Supplementary overview

This single Supplementary Information file contains Tables S1-S9 and Figures S1-S10 supporting the main Article. All analytical methods required to interpret the work are reported in the main manuscript. Machine-readable figure source data, regulator sensitivity results and complete statistical outputs are supplied separately as Supplementary Data 1-3. Original external-mapping sensitivity outcomes are superseded, and no corrected disease effect was estimated after calibration failed.

Supplementary Table S1 | Dataset roles and inferential units

Resource	Role	Biological unit	Active scope
GSE174188	Discovery and internal replication	Sample-cohort stratum; donor sensitivities	Disease-blind B_CONV/B_ASC analysis scaffold, composition and B_CONV transcription
GSE135779	Independent SLE replication	Donor	Broad conventional-B analogue; childhood primary
CollecTRI/ OmniPath	Curated regulator prior	Ranked genes within contrast	Prespecified eight-regulator, three-contrast family
MSigDB M5911	Orthogonal response prior	Ranked genes within contrast	Hallmark interferon-alpha response enrichment
GSE23307	Orthogonal perturbation support	Paired profile within donor	Descriptive IFN-beta response, n=2 donors

Supplementary Table S2 | Claim boundaries

Supported	Not supported
Disease-blind B_CONV/B_ASC analysis scaffold with quantified boundary sensitivity	Universally reproducible B_CONV/B_ASC taxonomy or stable hard naive, memory or atypical subtypes
Primary B_ASC contrast lacks statistical support	General B_ASC expansion in SLE
Replicated IFN/ISG remodeling within B_CONV	Genome-wide shared disease transcriptome
ULM STAT1/STAT2 activity concordant across three contrasts	Causal TF activation or direct binding
IFN-centred response evidence	A unique initiating IFN ligand in SLE

Supplementary Table S3 | Quantitative anchors and prespecified boundaries

Analysis	Result
Frozen-representation identity policy	minimum mapped ARI 0.990; minimum agreement 0.9998; minimum state-median Jaccard 0.991
End-to-end identity sensitivity	prespecified criterion not met; minimum mapped ARI 0.930; minimum agreement 0.9988; B_ASC median Jaccard 0.930 below 0.95 criterion
Boundary propagation	primary B_ASC odds-ratio range 0.896-0.967, all intervals include one; primary B_CONV IFN/ISG range 0.836-0.845, all intervals above zero
Primary B_ASC composition	43 controls and 47 source-defined managed SLE; odds ratio 0.947; 95% CI 0.636-1.410; P=0.787
GSE174188 primary IFN/ISG	effect 0.837; 95% CI 0.525-1.148; q=2.98 x 10^-6
GSE174188 donor-nonoverlap IFN/ISG	effect 1.086; q=3.61 x 10^-4
GSE135779 childhood IFN/ISG	effect 1.042; 95% CI 0.681-1.402; q=2.98 x 10^-6
Cross-dataset genome-wide agreement	4,410 genes; Spearman rho=0.026
M5911 enrichment	NES 3.187, 3.050 and 3.527
GSE23307 perturbation	donor effects 3.294 and 3.666; 12/12 genes positive in each

Supplementary Table S4 | Regulator-sensitivity summaries

a, Correlation-aware core-regulator sensitivity

Contrast	Regulator	Matched targets	Inter-gene correlation	CAMERA BH q	FRY BH q
GSE174188 primary	STAT1	98	0.0362	0.0263	6.043e-06
GSE174188 primary	STAT2	14	0.1225	0.1355	4.909e-05
GSE174188 donor-nonoverlap	STAT1	129	0.0209	0.0263	8.49e-06
GSE174188 donor-nonoverlap	STAT2	19	0.1191	0.0263	4.277e-07
GSE135779 childhood	STAT1	161	0.0301	0.0263	2.685e-05
GSE135779 childhood	STAT2	20	0.1223	0.03026	1.231e-05

CAMERA and FRY directions were positive in all six tests. CAMERA passed BH correction in five of six; the explicit exception was GSE174188 primary STAT2 (q=0.1355). FRY passed BH correction in all six.

b, IFN-overlap-depletion summary

Depletion	Method	Positive direction	Dedicated q<0.05	Main qualification
Frozen 12-gene IFN/ISG arm	ULM	6/6	6/6	Minimum slope retention 53.5%; all 95% confidence intervals above zero

Depletion	Method	Positive direction	Dedicated q<0.05	Main qualification
Frozen 12-gene IFN/ISG arm	CAMERA	6/6	5/6	Discovery STAT2 q=0.326
Frozen 12-gene IFN/ISG arm	FRY	6/6	6/6	Direction and corrected support retained
M5911	ULM	6/6	5/6	Discovery STAT2 retained 8/14 targets; slope 0.391, 95% CI -0.745 to 1.526; q=0.500
M5911	CAMERA	6/6	2/6	Broad attenuation across contrasts; discovery STAT2 q=0.623
M5911	FRY	6/6	5/6	Discovery STAT2 q=0.099

All 11 depleted ULM models retaining at least ten targets preserved direction in every leave-one-target analysis. These sensitivities support robustness beyond the narrow 12-gene arm but not independence from the broader interferon-response transcriptome.

Supplementary Table S5 | Selected figure source-data map

Figure	Evidence basis	Machine-readable source
Figure 1	Disease-blind identity stability, two-compartment adjudication and end-to-end B_ASC boundary	Figure1_source_data.csv
Figure 2	Sample-level composition in the 43-control/47-managed-SLE primary comparison	Figure2_source_data.csv
Figure 3	Raw-count pseudobulk transcription with explicit tested-gene symbols	Figure3_source_data.csv
Figure 4	Source-label-defined GSE135779 replication, gene-level coherence and required calibration boundary	Figure4_source_data.csv
Figure 5	Regulatory convergence, IFN-overlap-depletion ceiling and orthogonal response evidence	Figure5_source_data.csv
Supplementary Figure S4	End-to-end identity boundary and downstream propagation	Supplementary_Figure_S4_source_data.csv
Supplementary Figure S8	Corrected reference calibration and unresolved external transfer	Supplementary_Figure_S8_source_data.csv
Supplementary Figure S10	STAT1/STAT2 IFN-overlap-depletion sensitivity	Supplementary_Figure_S10_source_data.csv

Supplementary Table S6 | Reproducibility record

Component	Reader-accessible record
Main and supplementary figure data	Supplementary Data 1 with SHA-256 manifest

Component	Reader-accessible record
Complete statistical results	Supplementary Data 3 with 12 gene-level branches and 12 sanitized design matrices
Correlation-aware sensitivity	Supplementary Data 2
End-to-end identity and boundary propagation	Supplementary Data 3, `identity_robustness/`
Analysis code and decisions	`https://github.com/1209433622cz-maker/sle-bcell-remodeling`
Environment reconstruction	Pinned scientific and release environments documented in `REPRODUCIBILITY.md`
Version-specific archive	Zenodo https://doi.org/10.5281/zenodo.22151739; version-specific archive of the released analysis code, Source Data and statistical outputs

Supplementary Table S7 | Statistical tests and multiplicity families

Result family	Biological unit	Test or estimator	Sidedness	Multiplicity	Role
B_ASC composition	Sample-cohort stratum	Beta-binomial Wald	Two-sided	BH across three frozen base contrasts	Primary composition inference
Gene-level expression	Sample-cohort pseudobulk (GSE174188); donor pseudobulk (GSE135779)	edgeR robust quasi-likelihood F	Two-sided	BH within tested genes per contrast	Gene-level inference
Four frozen programs	Sample-cohort pseudobulk (GSE174188); donor pseudobulk (GSE135779)	OLS with HC3	Two-sided	BH across four programs per analysis	Program inference
TF-target activity	Ranked gene statistics	Signed-target slope t test	Two-sided	Global BH across 24 tests	Confirmatory regulatory evidence
STAT1/ STAT2 sensitivity	Ranked gene statistics	CAMERA and FRY	Positive-direction	Separate BH families of six per method	Correlation-aware sensitivity
STAT1/ STAT2 overlap depletion	Ranked gene statistics and pseudobulk counts	ULM, CAMERA and FRY	Two-sided ULM; positive-direction CAMERA/FRY	Separate BH family of six per branch and method	Post-freeze sensitivity
End-to-end identity reproducibility	Cell resample within technical library	ARI, AMI, mapping agreement, state Jaccard and recall	Not tested	Five unchanged threshold criteria	Disease-blind robustness boundary
Broad-state boundary propagation	Sample-cohort stratum	Frozen beta-binomial and TMM logCPM/HC3 models	Two-sided intervals	No new multiplicity family; same-data sensitivity	Assignment-uncertainty sensitivity
M5911	Ranked gene statistics	10,000-permutation preranked test	Positive-direction	Descriptive BH across three contrasts	Orthogonal support

Result family	Biological unit	Test or estimator	Sidedness	Multiplicity	Role
GSE23307	Paired donor	Mean paired log2(x+1) effect	Not tested	None; n=2	Descriptive perturbation

Supplementary Table S8 | Full statistical-results archive map

Directory	Contents	Files
gene_level_results	Complete filterByExpr gene results for GSE174188 and GSE135779 branches	12
sanitized_design_matrices	Direct-identifier-free analysis designs	12
composition	Frozen coefficients, contrasts, predictions, sensitivities, leave-one-out and diagnostics	8
transcription	Program, ranked-list, influence and concordance results	18
regulatory_and_orthogonal	ULM, target influence, CAMERA/FRY, M5911 and GSE23307 summaries	9
statistical_framework	Machine-readable test and multiplicity map	1
identity_robustness	Aggregate and replicate-level end-to-end identity metrics, boundary propagation results and integrity records	101
external_mapping_calibration	Corrected feature and parameter tables, full confidence-calibration family, aggregate selection diagnostics, provenance, arithmetic recount and publication boundary	20

Supplementary Table S9 | Reference-calibrated external mapping boundary

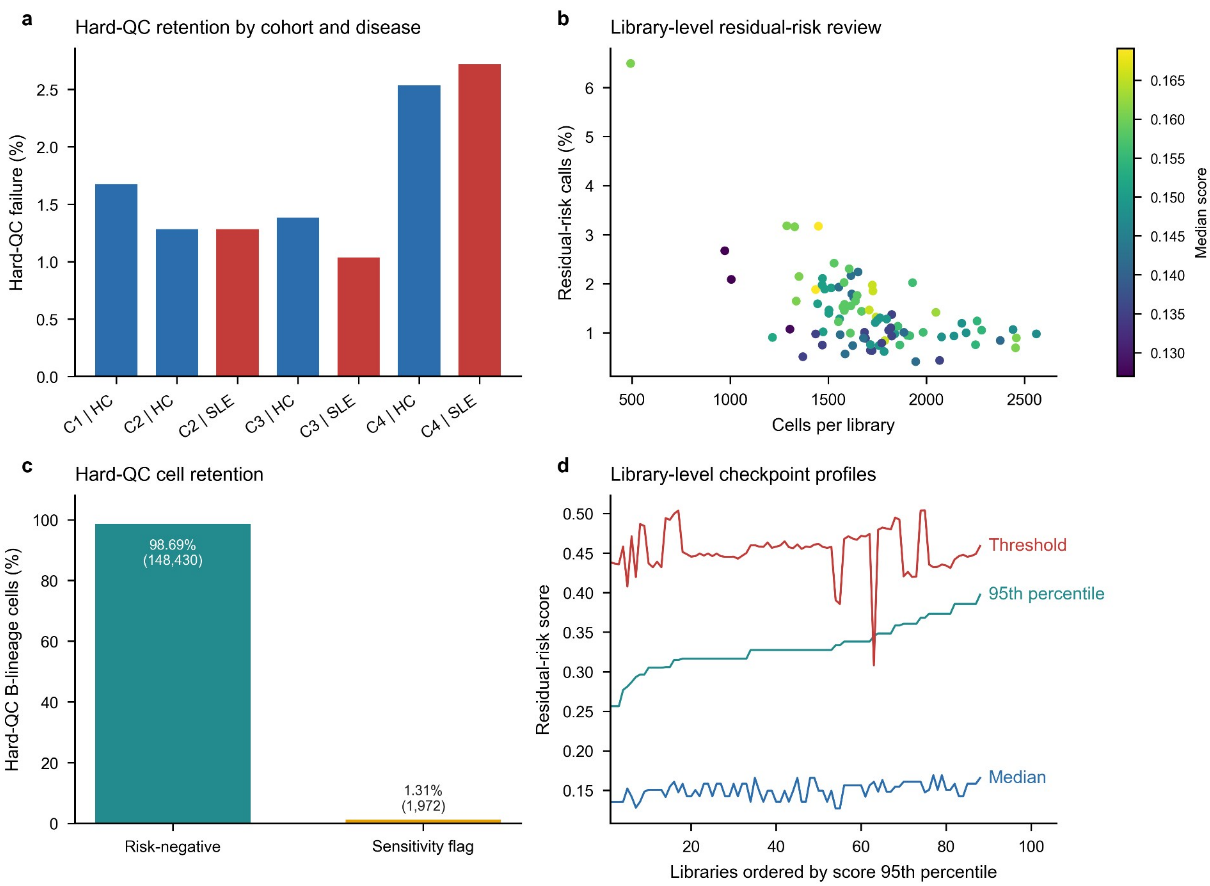
Component	Corrected diagnostic	Interpretation
External input	56 matrices; 363,083 cells	Complete source matrix coverage
External QC and selection	353,527 QC-passing cells; 36,630 B-lineage candidates	No source labels or disease fields parsed
Reference training	13,000 B_CONV and 1,300 B_ASC; 258 donors; 601 features	Both algorithms share reference and features
Normalization	Full-library log1p(CP10K) before feature subsetting in both datasets	Corrects reference feature-only denominator
Elastic-net calibration	Diagnostic threshold 0.95; coverage 0.941958; B_CONV precision 0.996450; B_ASC precision 0.885210	No eligible candidate: state precision must be at least 0.90 and coverage at least 0.80
Centroid calibration	Margin 0.117767; coverage 0.95; B_CONV precision 0.992374; B_ASC precision 1.0	Eligible, but cannot replace required primary mapper
Mean reference-fold balanced accuracy	Elastic net 0.960569; centroid 0.950293	Calibration diagnostics; not independent performance estimates

Component	Corrected diagnostic	Interpretation
Outcome policy	Corrected disease outcomes not estimated	No corrected effect, confidence interval, P or q value exists
Prior outcome exposure	Original sensitivity outcomes were known before correction	Post-outcome correction; not prospective validation
Evidence boundary	Primary GSE135779 replication remains source-label-defined	Superseded uncorrected outcomes excluded from supporting evidence

Full candidate calibration, donor-grouped cross-validation, reference normalization audit and corrected input/prediction hashes are retained with the analysis code. The original source-label-defined pseudobulk effect is a standardized HC3 model estimate; the unexecuted sensitivity would instead compare donor mean cell-level log1p(CP10K) scores. They are different estimands and must not be compared as effect attenuation. No new multiplicity family was evaluated after corrected calibration failed.

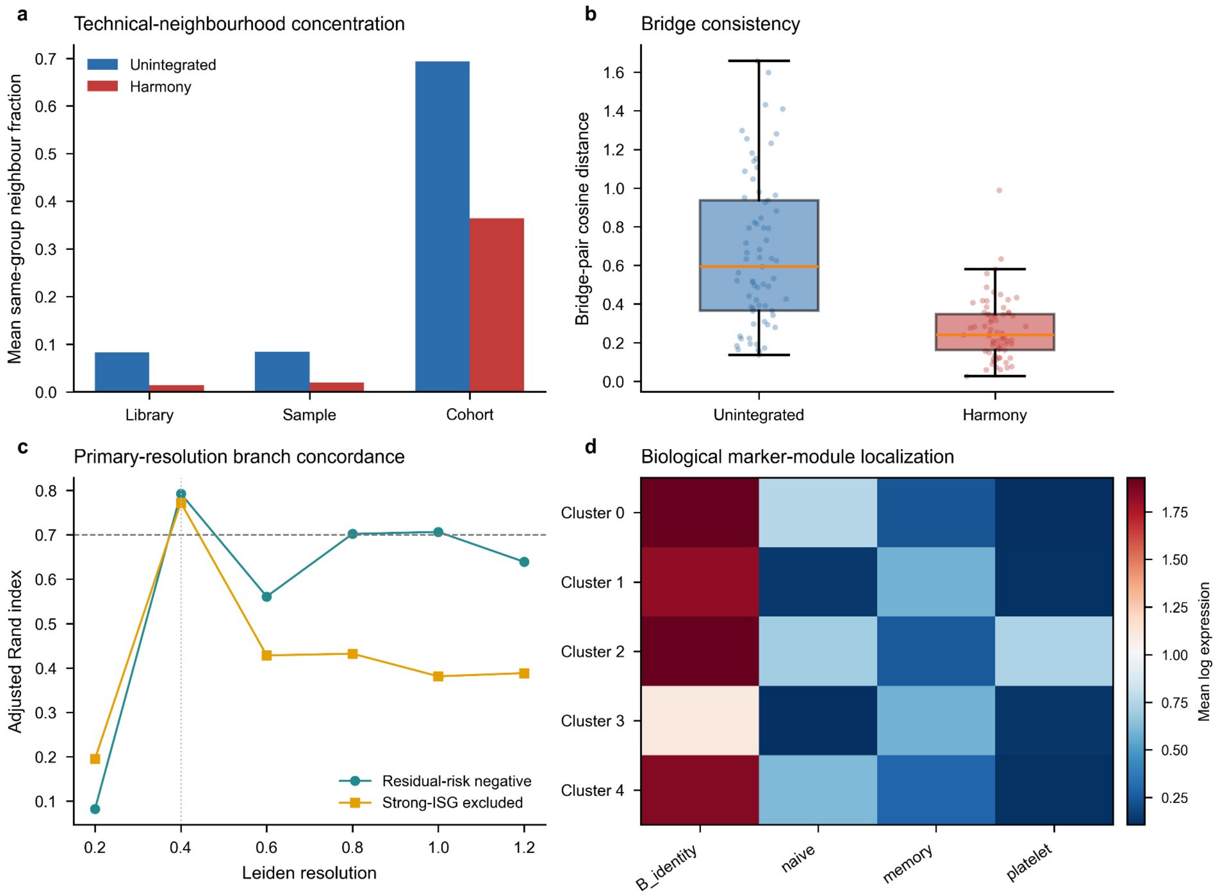
Supplementary Figure S1 | Source integrity and hard-quality-control diagnostics

a, Hard-QC failure fractions by processing cohort and disease group. **b,** Residual-risk call fraction versus cells per library, coloured by the library median score. **c,** Percent and exact number of retained risk-negative cells and sensitivity-only residual-risk calls among 150,402 hard-QC B-lineage cells. **d,** Median, 95th-percentile and threshold residual-risk scores for all 88 libraries.



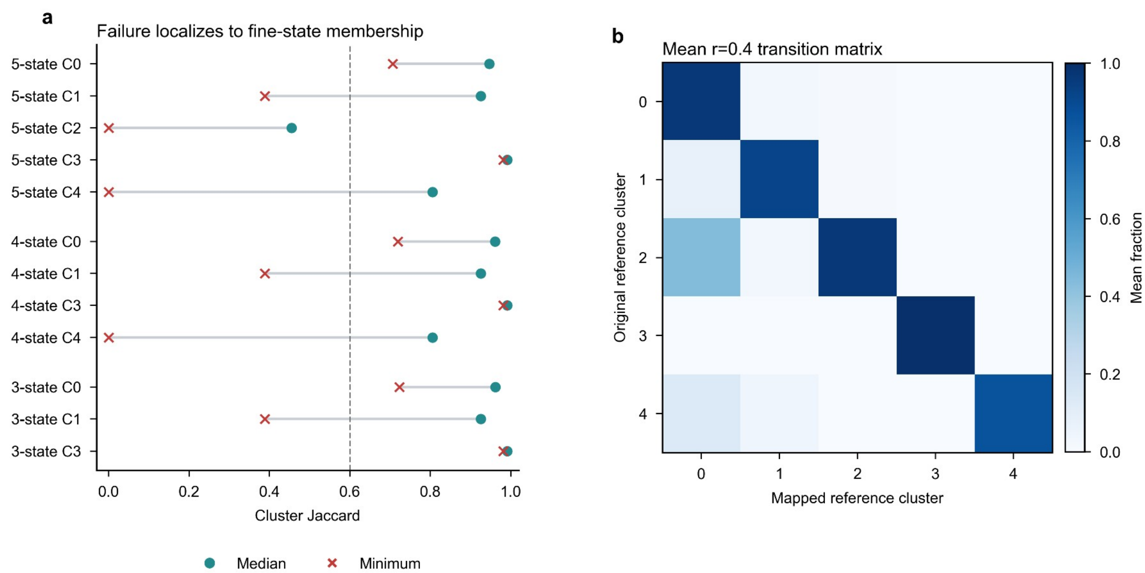
Supplementary Figure S2 | Representation and bridge diagnostics

a, Same-group neighbourhood concentration before and after Harmony adjustment. **b,** Cross-cohort bridge-pair cosine-distance distributions. **c,** Disease-blind branch concordance across Leiden resolutions for residual-risk-negative and strong-ISG-excluded branches; the vertical guide marks the primary resolution. **d,** Biological marker-module localization across the five fine clusters considered before stability adjudication.



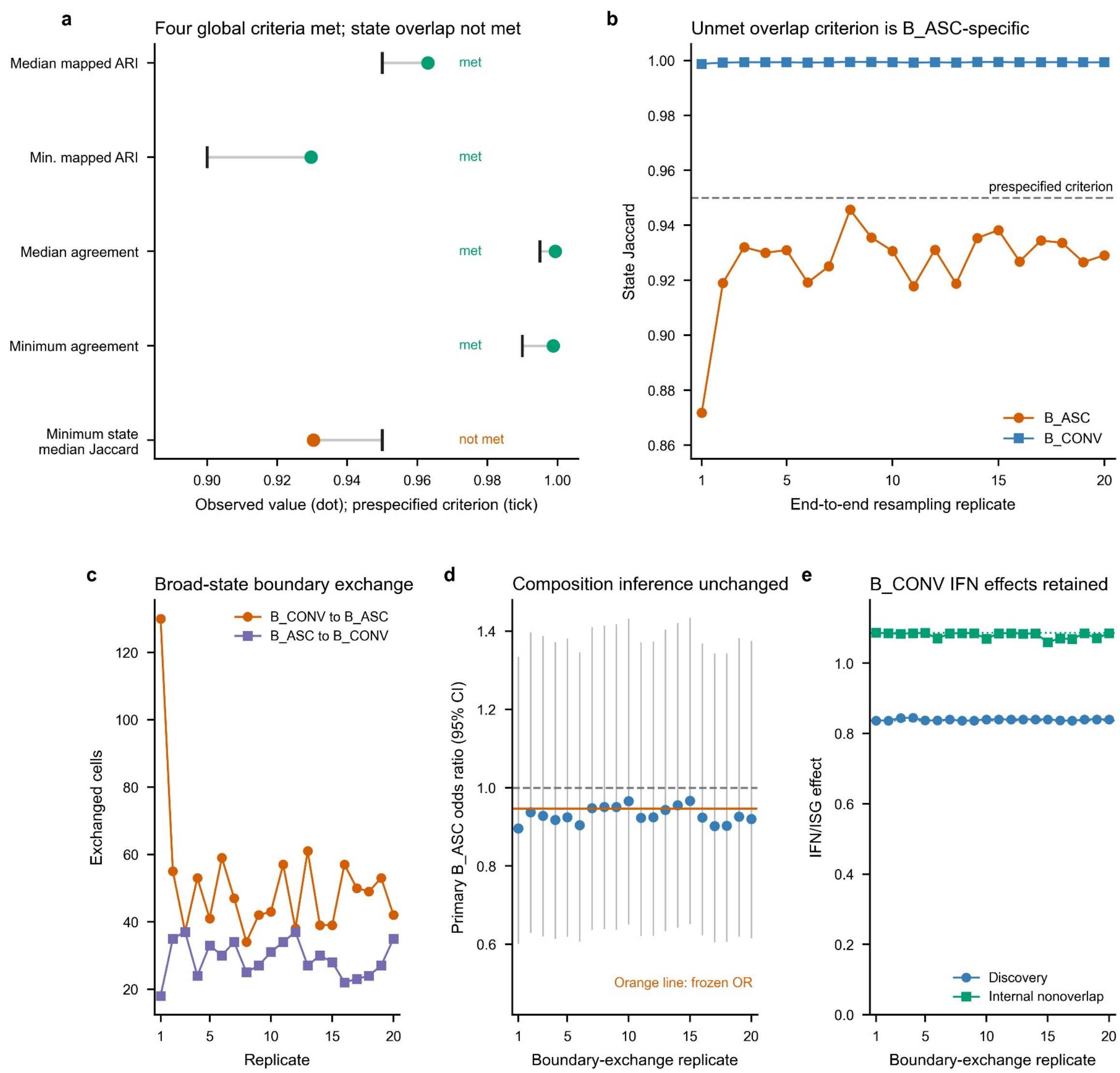
Supplementary Figure S3 | Fine-state failure and transition structure

a, Median and minimum cluster Jaccard values localize instability to fine-state membership across the five-, four- and three-state policies considered before broad adjudication. **b,** Mean transition matrix from original resolution-0.4 clusters to mapped reference clusters across frozen-representation resamples. These diagnostics explain the transition to the broad B_CONV/B_ASC analysis scaffold; broad-state pass criteria are shown in Fig. 1 and end-to-end reconstruction is shown in Supplementary Fig. S4. Highly variable genes, principal components and Harmony coordinates were not recomputed in this figure.



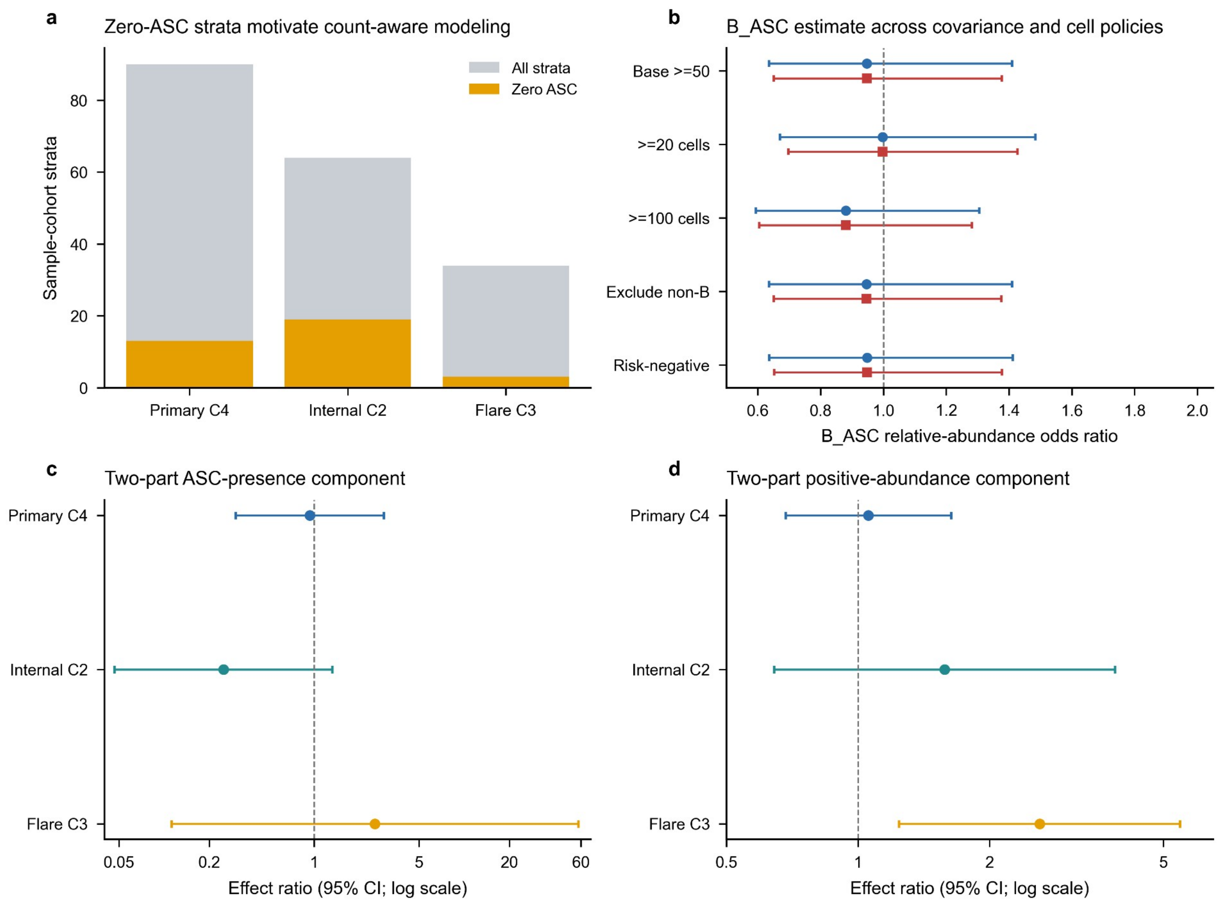
Supplementary Figure S4 | End-to-end reconstruction exposes a B_ASC-specific boundary without changing the disease conclusions

a, Observed values and prespecified criteria for five end-to-end two-compartment checks; four met their criteria and minimum state-median Jaccard did not. **b,** State Jaccard across 20 complete reconstruction replicates localizes the unmet overlap criterion to B_ASC, while B_CONV remains above 0.95. **c,** Counts of sampled cells exchanged across the B_CONV/B_ASC boundary. **d,** Primary B_ASC composition odds ratios and 95% confidence intervals after each observed boundary exchange; the dashed guide marks one and the orange line marks the frozen estimate. **e,** Primary and donor-nonoverlap B_CONV IFN/ISG effects after boundary-cell raw counts were propagated through frozen TMM logCPM and HC3 models; dotted lines mark the frozen effects. All propagation analyses reuse GSE174188 and quantify assignment sensitivity rather than independent replication.



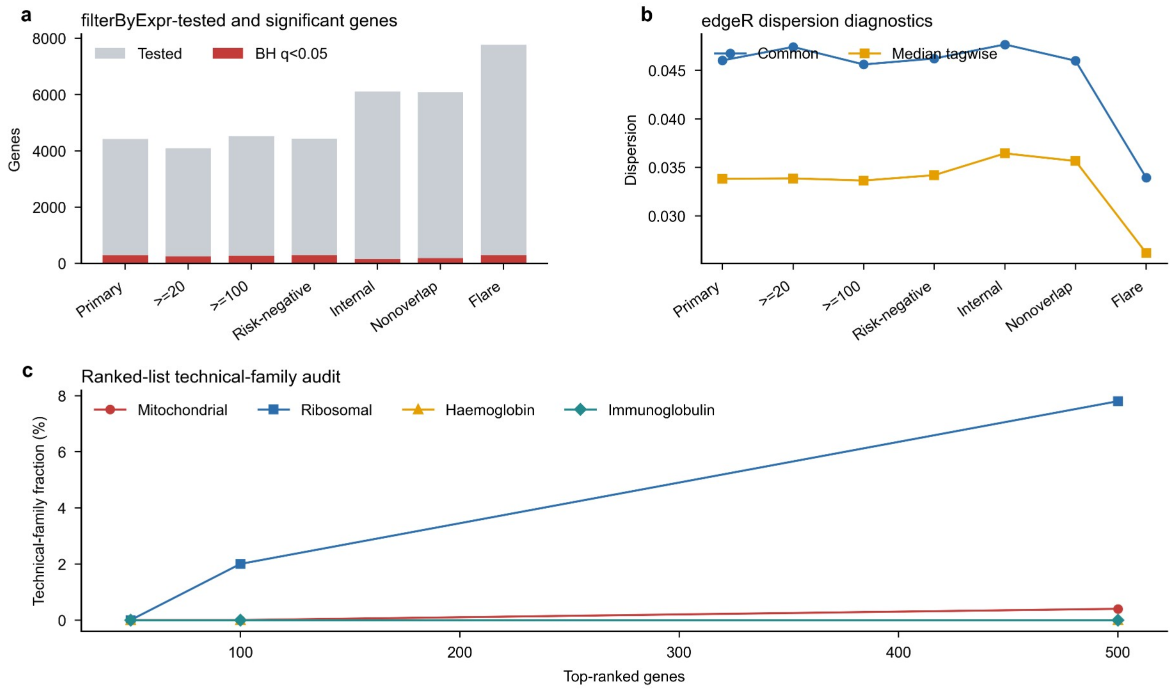
Supplementary Figure S5 | Composition-model diagnostics

a, Total and zero-ASC sample-cohort strata in the three base contrasts. **b,** Primary B_ASC odds ratios under support, explicit non-B and residual-risk policies using observed-information and HC1 covariance. **c,** Firth-logistic ASC-presence sensitivity. **d,** Positive-only abundance sensitivity using HC3 uncertainty. Panels c-d use logarithmic ratio axes with the null fixed at one; the two-part models are sensitivity analyses and do not replace the frozen beta-binomial primary model.



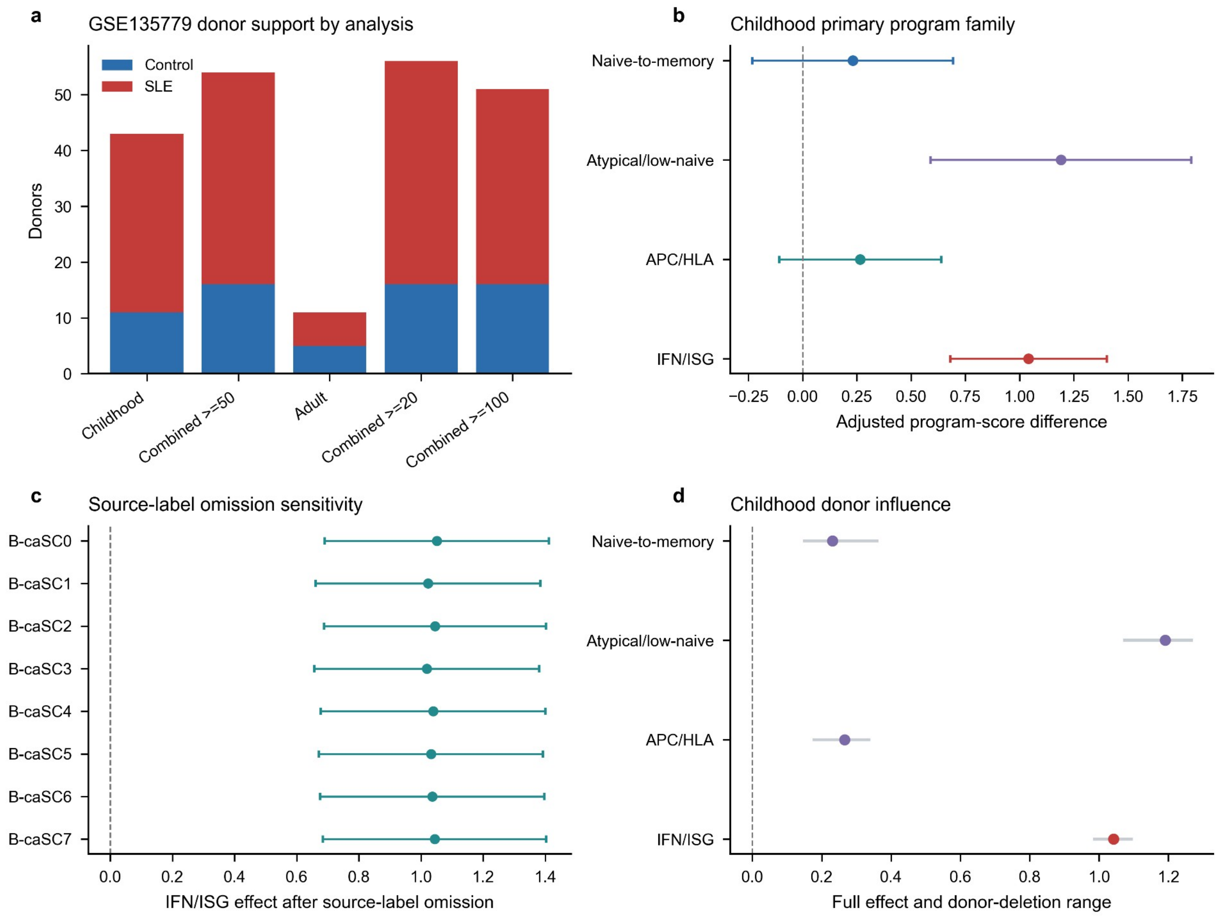
Supplementary Figure S6 | Pseudobulk and ranked-list diagnostics

a, Numbers of filterByExpr-tested and BH-significant genes across seven GSE174188 branches. **b,** Common and median tagwise edgeR dispersions. **c,** Mitochondrial, ribosomal, haemoglobin and immunoglobulin fractions among increasingly long primary ranked lists. IFN/ISG estimates across the frozen GSE174188 branches are owned by Fig. 3b and are not repeated here.



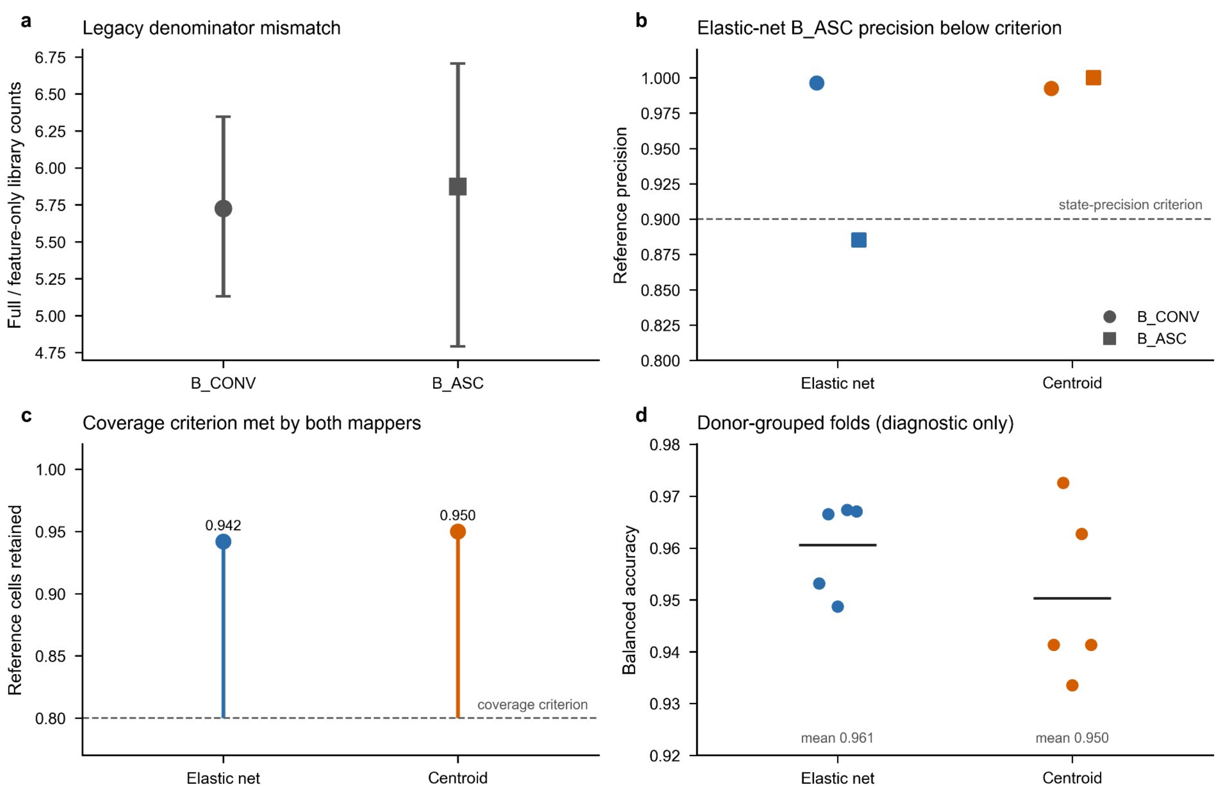
Supplementary Figure S7 | GSE135779 replication and robustness diagnostics

a, Control and SLE donor counts across the five GSE135779 models. **b,** Four-program childhood estimates. **c,** Childhood IFN/ISG estimates after omission of each source B-cell label. **d,** Full childhood program effects and ranges across donor-deletion analyses.



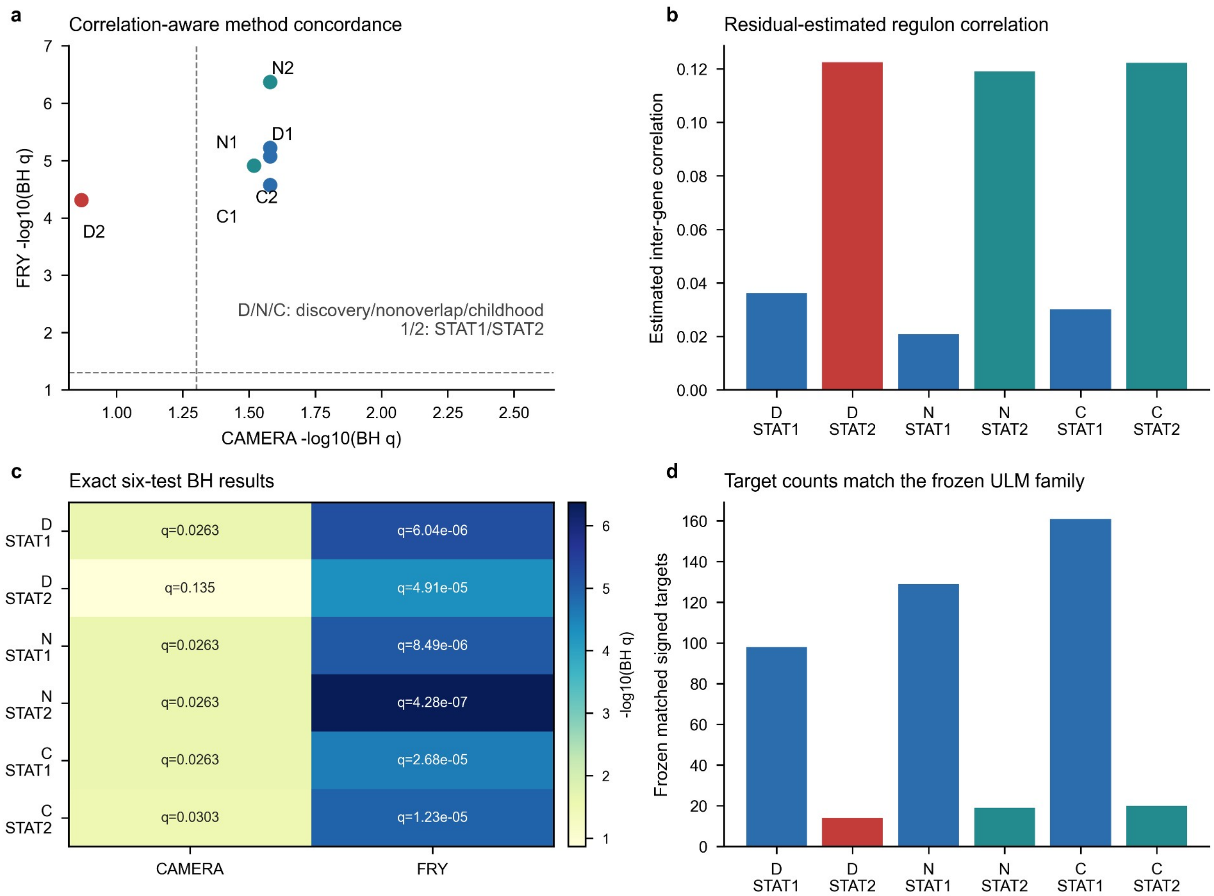
Supplementary Figure S8 | Reference calibration limits source-label-independent external transfer

a, Median and interquartile range of full-library divided by selected-feature counts among 13,000 B_CONV and 1,300 B_ASC reference training cells. This ratio quantifies the legacy pre-log scaling discrepancy; corrected mapping uses full-library denominators in both datasets. **b,** Per-state precision at the diagnostic elastic-net and eligible centroid thresholds. Mapper colour is held constant across panels b-d, marker shape distinguishes B_CONV from B_ASC, and the dashed line is the unchanged 0.90 state-precision criterion. **c,** Reference-cell coverage at those thresholds; the dashed line is the unchanged 0.80 criterion. **d,** Balanced accuracy in each of five donor-grouped reference calibration folds; black horizontal bars show fold means. No eligibility guide is drawn because balanced accuracy is diagnostic only. These folds select model and threshold parameters, not independent performance estimates. The elastic-net B_ASC precision failure prevents outcome access; centroid success is not a replacement analysis. No corrected external disease result is shown.



Supplementary Figure S9 | Correlation-aware regulator sensitivity

a, CAMERA and FRY six-test BH concordance for STAT1 and STAT2. Dashed guides indicate q=0.05. **b,** CAMERA residual-estimated inter-gene correlations. **c,** Exact CAMERA and FRY BH q values. **d,** Matched signed-target counts, which equal the frozen ULM family (STAT1: 98, 129 and 161; STAT2: 14, 19 and 20).



Supplementary Figure S10 | STAT1/STAT2 IFN-overlap-depletion sensitivity

a, ULM slopes and 95% confidence intervals after removal of the frozen 12-gene IFN/ISG positive arm. **b,** Corresponding estimates after removal of all 97 M5911 genes; discovery STAT2 retains eight targets and its interval crosses zero. Labels report remaining matched targets. **c,** Exact branch- and method-specific Benjamini-Hochberg q values across the six regulator-by-contrast tests. **d,** Percentage of frozen matched targets retained after each depletion. All method-level directions remain positive, but M5911 depletion produces substantial attenuation and does not support an overlap-independent interpretation.